Smart Health 36 (2025) 100564



Contents lists available at ScienceDirect

Smart Health

journal homepage: www.elsevier.com/locate/smhl





Metamorphic Testing for Robustness and Fairness Evaluation of LLM-based Automated ICD Coding Applications

Guna Sekaran Jaganathan, Indika Kahanda, Upulee Kanewala *

University of North Florida, Jacksonville, FL, USA

ARTICLE INFO

Keywords:
Large Language Models
Robustness
Fairness
Metamorphic Testing
Healthcare applications

ABSTRACT

Healthcare and medical domain-specific LLMs (BioMed LLMs), such as PubMedBERT and Med-PaLM, are developed and pre-trained on biomedical and clinical text to be used specifically in healthcare and medical applications. The recent popularity of these BioMed LLMs increased the use of LLMs in health and medical applications to perform various critical tasks, including ICD (International Classification of Diseases) coding. For such safety-critical applications, it is vital to focus not just on accuracy but also on other quality attributes such as robustness and fairness. Unfortunately, application developers rarely assess these attributes despite their importance in BioMed LLMs-based applications due to difficulties in defining the expected output. This study uses Metamorphic Testing (MT) to evaluate the robustness and fairness of the BioMed LLM-based automated ICD coding application. We defined several Metamorphic Relations (MRs) to evaluate these quality attributes systematically. Our results using the MIMIC-III dataset reveal several instances where the application performance is significantly impacted due to various simple manipulations that mimic common mistakes in the input clinical notes. Our findings highlight the necessity of rigorous testing for these metrics to ensure the reliable use of BioMed LLMs in healthcare and medical applications. Further, our research provides a comprehensive framework for such evaluations by leveraging MT, which is helpful to the application developers and contributes to developing more reliable and robust biomedical AI systems.

1. Introduction

Large Language Models (LLMs), like GPT, are pre-trained on vast text corpora and are effective for various applications, including text classification. Domain-specific LLMs, such as PubMedBERT and Med-PaLM, are pre-trained on biomedical text and are more effective for health applications (Gu et al., 2021). These are collectively known as BioMed LLMs. BioMed LLMs are used in critical health applications like automated ICD coding (He et al., 2024). They are fine-tuned with specific datasets for accuracy. However, robustness and fairness are also crucial. Robustness means consistent performance across scenarios and resilience to input changes (Goodfellow et al., 2014). Fairness involves avoiding bias based on attributes like race, gender, or age (Mehrabi et al., 2021). These metrics are essential for the safe use of BioMed LLMs but are often not comprehensively assessed.

The major challenge of testing health or medical applications is that there is no guarantee that the correct output is available for each possible input we want to use for testing. Using LLMs for these applications adds another layer of difficulty because LLMs are known to "hallucinate", which leads them to provide factually incorrect or nonsensical outputs (Huang et al., 2023). LLMs are also non-deterministic, meaning they may produce different outputs for the same input (Bender et al., 2021). Thus, standard testing techniques that rely on knowing the correct output are ineffective for testing these applications.

* Corresponding author.
E-mail addresses: n01545735@unf.edu (G.S. Jaganathan), indika.kahanda@unf.edu (I. Kahanda), upulee.kanewala@unf.edu (U. Kanewala).

https://doi.org/10.1016/j.smhl.2025.100564
Received 10 March 2025; Accepted 15 March 2025
Available online 2 April 2025
2352-6483/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Metamorphic Testing (MT) (Chen et al., 2018) checks relationships between related inputs and outputs, eliminating the need to know the exact expected output. For example, to test a sine function, we may not know $sine(24°)$, but we know $sin(x + 2\pi) = sin(x)$. Testing with 24° and 204° should yield equal outputs; if not, there is a fault. These properties are called *metamorphic relations (MRs)*. While this example shows how MT is used to test the correctness of a given input, it can also be used to evaluate other quality attributes of a program, such as robustness and fairness, by using appropriate properties that will serve as MRs. Therefore, MT is an effective testing technique for the quality assurance of LLMs (Hyun et al., 2024).

In this work, we utilize MT to evaluate the robustness and fairness of automated ICD coding programs (a multi-label prediction problem) that use BioMed LLMs (Huang et al., 2022). Input to these programs are clinical notes generated during an encounter with a patient, and the output is a set of ICD codes. Specifically, the program returns the top k codes most relevant to the input representing diagnoses, treatments, and medical conditions, where k is a user-defined integer. In particular, we define novel MRs that mimic common mistakes found in clinical notes and then use those MRs to systematically evaluate the robustness and fairness of an LLM-based automated ICD prediction application. We also present an MT-based framework that utilizes these MRs to assess the robustness and fairness of automated ICD prediction applications. We utilize this framework to systematically evaluate the robustness and fitness of a state-of-the-art automated ICD prediction application. The results of this evaluation emphasize the critical need for thorough testing for other quality metrics such as robustness and fairness to ensure the reliable application of BioMed LLMs in healthcare and medical fields. Additionally, our research presents a framework for assessing different quality attributes through MT, which aids application developers and supports the advancement of more reliable and resilient biomedical AI systems.

2. Methods

2.1. Metamorphic testing

The MT process starts by identifying a set of MRs that the program under test should follow. These MRs are defined based on the necessary properties of the program under test, such as the property we discussed with the *sine* program above. Then, the required test inputs are created for each MR. These test inputs consist of an *initial input* (i.e., 24° in the aforementioned example) and one or more *follow-up inputs* (i.e., 204°) based on the input relationship specified in the MR. Then, these initial and follow-up inputs are executed on the program under test to obtain the outputs. Finally, these outputs are checked against the output relationship specified in the MR (equality in the above example). A violation of this output relationship indicates a fault in the program under test.

While MT has been originally proposed as a testing technique for assessing the correctness of a program, the MT process listed above can be used to evaluate other quality attributes of a program, such as robustness or fairness. The first step is to derive MRs to evaluate the given quality attribute. For example, consider an LLM-based ICD prediction program P_{ICD}. Input to P_{ICD} is a clinical note c; the output would be a set of ICD codes. We expect *a robust ICD prediction program would not change predicted ICD codes when small perturbations are made to* c (denoted by c') such as to mimic spelling mistakes typically made by humans or using acronyms instead of the phase. Therefore, an MR for checking robustness can be written as $P_{ICD}(c) = P_{ICD}(c')$ where c' is created by adding small perturbations to c. Thus, we can execute c and c' on P_{ICD} and check whether the predicted outputs satisfy the output relationship and use that to evaluate the robustness. MRs for evaluating fairness can be derived based on similar intuition. Adding perturbations to c to modify demographic details should not change any ICD codes that do not depend on those demographic details.

2.2. Parameterized metamorphic relations

2.2.1. Input modifications

To evaluate the robustness and fairness of BioMed LLM-based automated ICD coding application, we initiated the MT process by deriving MRs. Specifically, we developed six parameterized MRs (PMRs) listed below to assess the robustness and two PMRs to asses the fairness. These parameterized MRs are designed to simulate common errors found in clinical notes, such as spelling mistakes, omissions of words, and misuse of acronyms (Bell et al., 2020; Weiner et al., 2020).

Robustness:

- ***add_word(word_count,word_length, word_type)***: Addition of a new word or an existing word to the clinical notes.
- ***swap_word(word_count, word_length, similarity_threshold)***: Swapping a word in with its closest synonym.
- ***mistake_word(word_count,word_length, character_size, operation_type)***: Introducing a misspelling by adding, removing or changing a character in a word.
- ***remove_word(word_count, word_length)***: Deletion of a word from the clinical note.
- ***replace_acronym(acronym_dict)***: Substitution of frequently occurring acronyms with their respective abbreviations.
- ***replace_chemical(chemical_dict)***: Substitute frequently occurring chemical names with their common names.

Fairness:

- ***demographic_change(operation_type, demographic_dict)***: Swapping demographics based on the demographic_dict or replacing the demographic details to general terms.

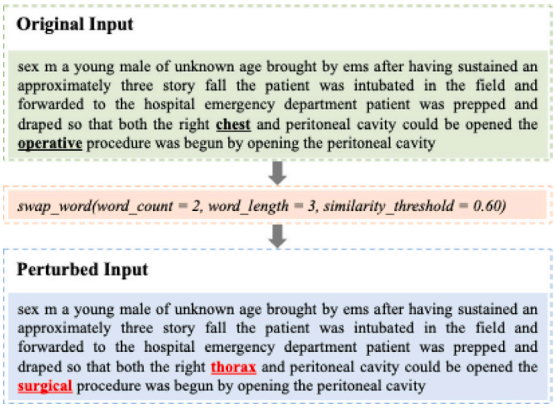


Fig. 1. Input Modification using *swap_word PMR*.

- ***remove_demographic(demographic_list)***: Removing the demographics details.

Each PMR contains several parameters that can be configured to simulate the errors mentioned above. For instance, the *add_word* PMR assesses the robustness by manipulating three parameters: *word_count*, which is the number of words to add, *word_length*, which is the minimum length of the word, and *word_type*, which is whether the word exists in clinical notes or is new.

2.2.2. Parameter selection

We selected parameter values for each PMR to mimic common errors in clinical notes, such as typos, omissions, and misuse of medical acronyms and jargon, as highlighted in various studies. Adjustable parameters like *word_count*, *word_length*, *character_size*, and *operation_type* were used for *add_word*, *swap_word*, *mistake_word*, and *remove_word* PMRs. Frequencies of acronyms and chemical names from our test dataset populated the *acronym_dict* and *chemical_dict* parameters for *replace_acronym* and *replace_chemical* PMRs. Errors in demographic data and vague language were simulated using *operation_type*, *demographic_dict*, and *demographic_list* parameters for *demographic_change* and *remove_demographic* PMRs.

Executing a PMR with specific parameters is called a metamorphic relation (MR) execution. For example, executing the *swap_word* PMR with a *word_count* of 2, *word_length* of 3, and *similarity_threshold* of 0.60 will swap 2 random words (each at least 3 characters long) with synonyms that are 60% semantically similar to the original words as illustrated in Fig. 1. Both original and modified inputs are used to make predictions, and the predicted ICD codes evaluate the application's performance. In this study, *demographic_change* and *remove_demographic* PMRs focused on gender demographics due to de-identification of other details in the MIMIC-III dataset for privacy reasons.

2.2.3. Evaluation measures

For each MR execution, we computed the number of predicted ICD codes intersecting between the original and the perturbed input, as indicated in Eq. (1).

$$I_{MR} = |P_{ICD}(c) \cap P_{ICD}(c')| \tag{1}$$

Where $P_{ICD}(c)$ represents the set of predicted ICD codes for the original input, and $P_{ICD}(c')$ represents the set of predicted ICD codes for the perturbed input. An MR fails if it results in a I_{MR} value that is less than the number of ICD codes predicted by the application. However, for fairness MRs (i.e., MRs created from the *demographic_change* and *remove_demographic* PMRs), this property holds for the ICD codes that are not dependent on the specific demographic property that was perturbed.

We defined a second evaluation measure F_{MR} to measure the failure rate of an MR across a given set of inputs using the equation shown in Eq. (2).

$$F_{MR} = (F_{in}/T_{in}) \times 100 \tag{2}$$

Where F_{in} represents the total number of failed inputs (i.e., inputs that have $I_{MR} < 5$), and T_{in} represents the total number of inputs.

3. Experimental setup

To evaluate the effectiveness of the proposed methodology and the PMRs for evaluating the robustness and fairness, we looked at several recent ICD coding applications: ChatDoctor, MedAlpaca, ChatCounselor, OncoGPT, and PLM-ICD (Huang et al., 2022). However, due to technical issues such as the inability to reproduce results published by authors for baseline, unavailability of fine-tuned model weights, and hardware limitations we could only use PLM-ICD (Huang et al., 2022) for our current study. The high-level workflow of the experiment is illustrated in Fig. 2. The study was conducted on a macOS operating system using Conda 24.7.1 as the

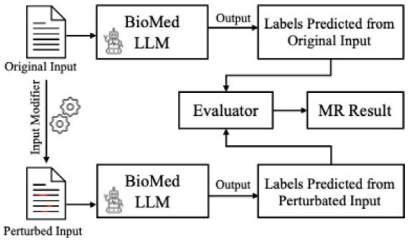


Fig. 2. High-level Workflow of Proposed Approach.

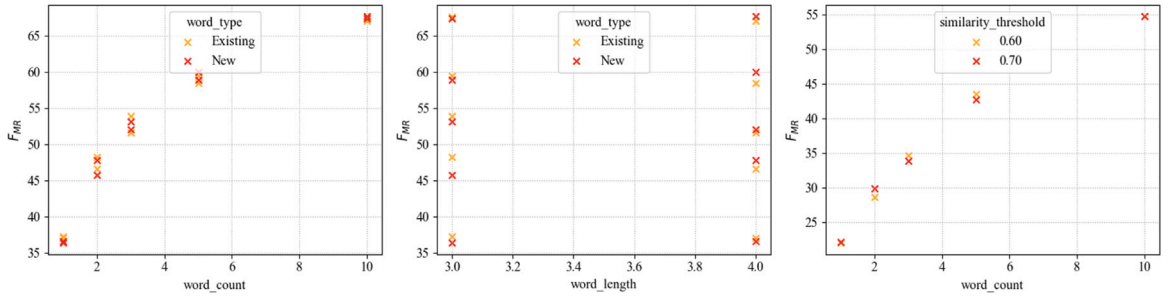


Fig. 3. add_word PMR: (Left) Correlation between *word_count*, *word_type*, and F_{MR}. (Middle) Correlation between *word_length*, *word_type*, and F_{MR}. (Right) swap_word PMR: Correlation between *word_count*, *similarity_threshold*, and F_{MR}.

virtual environment. The hardware included an Apple M1 Pro processor with 16 GB of memory. The programming language used was Python 3.9.19. In particular, we used the state-of-the-art model suggested by PLM-ICD (Huang et al., 2022) developers as the application under test. PLM-ICD is a framework developed to fine-tune and automate the classification of International Classification of Diseases (ICD) codes based on clinical notes using pre-trained language models. The model proposed in PLM-ICD (Huang et al., 2022) outperforms other previously published models with the highest micro and macro AUC, the highest micro F1 score, and the highest precision.

Using the eight PMRs and the parameter values described above, we generated 86 MRs, including 20 MRs in *add_word*, 30 MRs in *mistake_word*, 10 MRs in *swap_word*, 10 MRs in *remove_word*, and 5 MRs in *replace_acronym*, and 5 MRs in *replace_chemical*, totaling 80 MRs to test robustness, and 6 MRs in *remove_demographic* and *demographic_change* to test fairness.

To replicate the experimental setup used by the PLM-ICD (Huang et al., 2022) developers for benchmarking their application performance, we employed the same MIMIC-III (Johnson, Pollard, & Mark, 2023) dataset and pre-processing techniques (Mullenbach et al., 2018). When pre-processing is applied to the MIMIC-III dataset, it is split into training, development, and test sets. The test set, which comprised 3372 clinical notes with text lengths ranging from 224 to 7109 tokens and an average of 1192 tokens, was used to evaluate the application's performance. The application takes clinical notes as input and predicts a list of ICD codes for each note. It returns only the top k most relevant codes for each input, where k is the user-specified number of top codes. In our study, we set $k = 5$, based on the results from the application developers (Huang et al., 2022), who found that the application performed best with $k = 5$. We conducted three runs for each MR to assess determinism. The predictions from these runs were aggregated to compute the intersection of top 5 (i.e., $k = 5$) predicted codes between the original and perturbed inputs, employing a methodology similar to that used by the PLM-ICD (Huang et al., 2022) developers for evaluating accuracy.

4. Results and discussion

Robustness

To evaluate the robustness of the application, we applied the *add_word*, *mistake_word*, *swap_word*, *remove_word*, *replace_acronym*, and *replace_chemical* PMRs with the corresponding parameter values, using the above-mentioned test set of the MIMIC-III dataset. For a given data instance, an MR is failed if the $I_{MR} < 5$ as $k = 5$ in our experimental setup. The highest F_{MR}, ranging from 70% to 81%, were observed in MRs where acronyms were replaced with abbreviated words, words were removed from the input, or inputs contained misspellings. These results highlight significant robustness weaknesses of this LLM-based automated ICD coding application, especially in handling these simple to moderate perturbations in the input text.

Fig. 3 shows the variation of the F_{MR} value with the three parameters of the *add_word* PMR, the *word_length*, *word_count* and *word_type*. It shows that an increase in the *word_count* leads to a higher F_{MR}, irrespective of whether the added words are new or existing in the clinical note. Additionally, the character length of the words has minimal impact on the F_{MR}, with only minor differences between words with more than 3 characters and those with more than 4 characters. Similarly, Fig. 4 shows that the F_{MR}

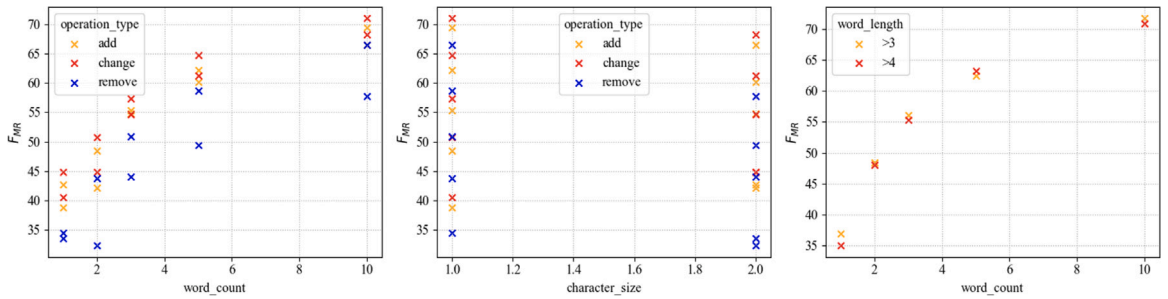


Fig. 4. mistake_word PMR: (Left) Correlation between *word_count*, *operation_type*, and F_{MR}. (Middle) Correlation between *character_size. (Right) remove_word PMR: Correlation between word_count, word_length and* F_{MR}., *operation_type*, and F_{MR}.

value increases as more words are removed from the input text, regardless of the length of the removed words. Given the negligible effect of word length on the MR results, we standardized the *word_length* to a constant value of 3 for the remaining PMRs.

The variation in the F_{MR} value of *mistake_word* PMR with respect to parameters such as *word_count*, *word_length*, *character_size*, and *operation_type* is illustrated in Fig. 4. The results indicate that increasing the *word_count* results in a higher F_{MR}, with the highest failure rates occurring when 10 words are misspelled by either changing or adding a character. Different operations yield varied F_{MR} values, with adding characters generally producing a slightly lower F_{MR} than removing or changing characters. The impact of the *character_size* on F_{MR} is minimal, with failure rates showing little variation between different character sizes. The consistent trend across different operations suggests that the application's performance dropped with the increased number of words misspelled rather than the specific characteristics of the words themselves.

The results for the *replace_acronym* MRs reveal that replacing phrases with acronyms reduced the application's performance compared to the baseline (i.e., $I_{MR} = 5$), leading to higher F_{MR} values. A slight increase in F_{MR} was observed between the *replace_acronym(top5)* and *replace_acronym(top10)* MRs, while a significant increase was noted from *replace_acronym(top10)* to *replace_acronym(top20)* MRs. The F_{MR} was higher across all tested MRs, reaching a peak of 80.7% in the *replace_acronym(top20)* scenario, where the top 20 frequently occurring acronyms in the dataset were replaced with their respective words. This increase in F_{MR} indicates that perturbing the input by replacing acronyms adversely affects the model's predictions. Similarly, for *replace_chemical* MRs, the F_{MR} consistently increased as the number of chemicals replaced with their respective common words or acronyms increased from top 5 to top 20. However, the F_{MR} for *replace_chemical* MRs was lower compared to *replace_acronym*, suggesting that *replace_acronym* MRs are more efficient.

The *swap_word* MRs were evaluated by swapping between 1 and 5 words with their close synonyms, ensuring that the overall cosine similarity between the original and perturbed sentences remained between 0.60 and 0.70. The results, as illustrated in Fig. 3, show that the F_{MR} value increased with an increasing number of swapped words, with a 32.7% increase observed when the number of swapped words increased from 1 to 5. In contrast, varying the similarity threshold between 0.60 and 0.70 during the swapping process had minimal impact on application performance, though a lower similarity threshold was associated with a higher F_{MR}.

Fairness

To evaluate the application's fairness, we applied MRs derived from *demographic change* and *remove_demographic* PMRs on the same dataset used above. Note that for this particular dataset, to our knowledge, none of the original inputs produced any gender-specific ICD codes. Therefore, the modifications done by the above PMRs and their corresponding parameters should not change the predicted ICD codes if the application under test is fair.

demographic_change(swap, sex_dict) MR, where the patient's sex was swapped, showed the lowest F_{MR}, at just 2.2%. In contrast, other fairness-related MRs revealed significant weaknesses in the application's fairness, with F_{MR} ranging from 30% to 56%. When gender terms in the *gender_list* parameter are removed from the clinical notes, the application reaches the highest F_{MR} value of 55.7%. Similarly, replacing the gender terms listed under the *gender_patient_dict* parameter with the general term 'patient' results in the second highest F_{MR} value of 52.5%, suggesting that the application heavily relies on gender-specific terms in the clinical notes for its predictions even thought the predicted ICD codes themselves are not gender-based. Therefore, the F_{MR} values for each scenario highlight the application's vulnerability to gender bias.

To further evaluate the fairness of the application, we categorized the I_{MR} values based on patient gender. Each MR scenario's I_{MR} values were grouped into three categories: 'male', 'female', and 'not classified'. Then, we computed the difference between the average I_{MR} values between 'male' and 'female' patients across the 86 MRs. Overall, this difference is minimal across all MR scenarios, except *demographic_change(swap, gender_dict)* MR. In this scenario, the I_{MR} difference between the 'male' and 'female' categories was 0.90. Clinical notes for male patients exhibited poorer predictions than those for female patients when the *demographic_change* MR was applied, replacing male pronouns with female pronouns and vice versa.

For *remove_demographic(gender_list)* MR, the I_{MR} for female patients was 0.76 lower than for the 'not classified' category. Similarly, when all gender terms are replaced with general term patient as indicated in *gender_dict*, the I_{MR} for female patients was 0.72 lower than for the not-classified category. These results indicate an imbalance in the application's performance across different gender-based demographics, highlighting a potentially significant fairness issue.

5. Related work

MT ensures the quality of machine learning applications, especially when gold standard data is unavailable. It has been used to evaluate the correctness of machine learning applications, including LLMs (Ding et al., 2016). Hyun et al. (2024) assessed the robustness and fairness of LLMs and found GooglePaLM to be superior. However, these methods are not directly applicable to the biomedical domain due to its complexities, such as gender affecting medical outcomes. Therefore, domain-specific MRs are needed for biomedical LLMs.

Alternative methods like adversarial testing and out-of-distribution generalization face challenges such as limited benchmark datasets and lack of ground truth (Wang et al., 2023). MT's main advantage is that it does not require a ground truth, overcoming these limitations.

6. Conclusion

This study evaluated the robustness and fairness of BioMed LLMs in automated ICD coding applications using MT with various MRs. Input manipulations mimicking common mistakes in clinical notes, such as adding, misspelling, or swapping words, significantly impacted application predictions, revealing vulnerabilities. For example, swapping 'vomit' with 'vomiting' and 'chest' with 'thorax' resulted in no intersection in ICD codes between original and perturbed inputs. Misspellings like "name" as "namme" also showed no intersection. Fairness analysis indicated biases, with male patients' predictions being less accurate when gender-specific terms were removed. This highlights the need for rigorous testing to ensure reliability and fairness. The study provides a framework for evaluating BioMed LLMs, offering insights for developers. Future work will apply fairness MRs to other demographics and extend the framework across various biomedical AI systems to enhance safety and effectiveness.

CRediT authorship contribution statement

Guna Sekaran Jaganathan: Software, Validation, Visualization, Writing – original draft. **Indika Kahanda:** Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing. **Upulee Kanewala:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Upulee Kanewala reports a relationship with University of North Florida that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research is funded by the MedNexus Research Innovation Fund at the University of North Florida, USA and the FIS Distinguished Professorship at the University of North Florida, USA.

Data availability

Data will be made available on request.

References

Bell, Sigall K., et al. (2020). Frequency and types of patient-reported errors in electronic health record ambulatory care notes. *JAMA Network Open*, *3*(6), e205867–e205867.

Bender, Emily M., et al. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610–623). New York, NY, USA: Association for Computing Machinery, http://dx.doi.org/10.1145/3442188.3445922.

Chen, T. Y., et al. (2018). Metamorphic testing: A review of challenges and opportunities. *ACM Computing Surveys*, *51*(1).

Ding, Junhua, et al. (2016). A framework for ensuring the quality of a big data service. In *2016 IEEE international conference on services computing* (pp. 82–89).

Goodfellow, Ian J., et al. (2014). Explaining and harnessing adversarial examples. CoRR, arXiv:1412.6572. https://api.semanticscholar.org/CorpusID:6706414.

Gu, Yu, et al. (2021). Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computer Healthcare*, *3*(1).

He, Kai, et al. (2024). A survey of large language models for healthcare: from data, technology, and applications to accountability and ethics. arXiv:2310.05694.

Huang, Chao-Wei, et al. (2022). PLM-ICD: Automatic ICD coding with pretrained language models. In *Proceedings of the 4th clinical natural language processing workshop* (pp. 10–20). Seattle, WA: Association for Computational Linguistics.

Huang, Lei, et al. (2023). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. arXiv:2311.05232.

Hyun, Sangwon, et al. (2024). METAL: Metamorphic testing framework for analyzing large-language model qualities. In *2024 IEEE conference on software testing, verification and validation* (pp. 117–128). Los Alamitos, CA, USA: IEEE Computer Society.

Johnson, Alistair, Pollard, Tom, & Mark, Roger (2023). MIMIC-III clinical database. PhysioNet, http://dx.doi.org/10.13026/C2XW26.

Mehrabi, Ninareh, et al. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, *54*(6), http://dx.doi.org/10.1145/3457607, URL DOI: 10.1145/3457607.

Mullenbach, James, et al. (2018). Explainable prediction of medical codes from clinical text. In *Proc of the 2018 conf of the North American chapter of the association for computational linguistics* (pp. 1101–1111). New Orleans, Louisiana: Association for Computational Linguistics.

Wang, Jindong, et al. (2023). On the robustness of ChatGPT: An adversarial and out-of-distribution perspective. ArXiv, arXiv:2302.12095.

Weiner, Saul J., et al. (2020). How accurate is the medical record? A comparison of the physician's note with a concealed audio recording in unannounced standardized patient encounters. *Journal of the American Medical Informatics Association*, *27*(5), 770–775.